

Section 3.3: Storage Media Introduction

In this section you will be introduced to the 3 main types of storage media:

- **Magnetic storage media**
- **Optical storage media**
- **Solid state storage media**

We also discuss **storage media capacity**, including **kilobytes**, **megabytes**, **gigabytes** and **terabytes**.

Key Concepts of this section:

- # Know the difference between **storage devices** and **storage media**.
- # Be able to **name** the different **storage capacity units** (from smallest to largest).

Storage Media (a quick introduction)

Storage media is the term given to the devices that are used to **hold data**.

Note: Storage media is **NOT** the same thing as a storage device.

- **Storage media** = the part that holds the data
- **Storage device** = the machine that reads/writes data to and from the media

Storage media is '**permanent storage**' and will **hold data** even when the computer is **powered off**.

Remember: **Permanent storage devices:**
Devices that hold data without power and are known as '**non-volatile**'

Temporary storage devices:
Devices that lose data without power and are known as '**volatile**'.

Storage media can either be:

- **Internal** - inside the computer
- **External** - plugged into the computer from outside (e.g. usb stick)

There are 3 different types of storage media and they all store data in different ways. These are summarised below:

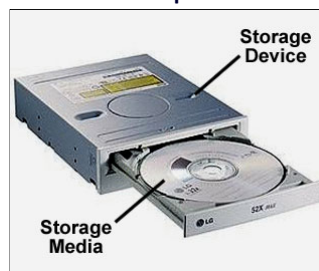
Storage Media	Summary
1. Magnetic media	• Holds data magnetically • Surface area of the media are magnetised so that they can hold 'bits' of data • The device that reads the media can also read the bits held on the surface area.
2. Optical media	• Holds data by burning 'dots' onto the surface with a high powered laser • The device that reads the media can read the data by shining a lower powered laser over the dots .
3. Solid-state media	• Holds data on devices that have no moving parts.



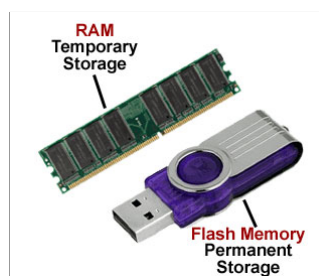
Key Words:

Storage device , Storage media, Internal, External, Magnetic, Optical, Solid-State

Examples:



The CD is the 'media' (where data is held)
The CD Drive is the 'device' (reads data on media).



There are 2 types of computer storage - Permanent and Temporary.

NOTE:

All 3 types of media are covered in more detail in the next 3 sections of this unit.

Storage Capacity

Storage capacity is the term given to the **maximum amount of data** that can be **stored** on the media.

Storage amounts are measured in '**bytes**' and there are **8 bits in 1 byte**.

Bits are short for '**binary digits**'. This is the smallest unit of data that can be stored. Each '**bit**' is represented as a **binary number**, either **1 or 0**.

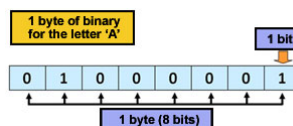
A single letter on a keyboard takes up 1 byte of storage (8 bits).



Key Words:

Bit, Byte, Kilobyte, Megabyte, Gigabyte, Terabyte

Examples:



There are 8 bits (binary digits) in 1 byte of data.
(click to zoom)

Computers read and process **everything** in the form of **binary numbers**. Each binary number takes up **1 bit of storage**.

For example: The letter **C** looks like **01000011** to a computer. There are 8 binary numbers.... so '**C**' takes up **8 bits** or **1 byte** of storage.

Click [here](#) for a text to binary converter to see what other keyboard characters look like in binary.

Storage Capacity Units:

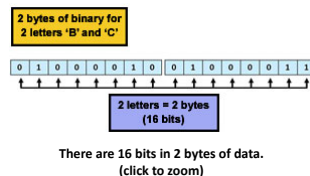
We refer to the storage capacities of different media in the following units:

Unit	Abbreviation	Information	
Bit		Smallest unit of data (either a 1 or a 0)	Smallest  Largest
Byte		1 byte = 8 bits	
Kilobyte	KB	1 KB = 1024 bytes	
Megabyte	MB	1 MB = 1024 Kilobytes	
Gigabyte	GB	1 GB = 1024 Megabytes	
Terabyte	TB	1 TB = 1024 Gigabytes	

Storage media all have different storage capacities.

For example: **Hard disk drives** have capacities in the **terabytes**.

Floppy disk drives can only hold **1.44MB** of data.



Floppy disks can't store very much data (about 1.44 MB).



CD's can hold about 800 MB of data.

Storage Capacity Examples:

The table below gives you some real-world examples of approximately how much data we could store on storage media with different capacities:

Storage Media Capacity	How much Data could be Stored
1 kilobyte	1 thousand characters (half a page of text)
1 megabyte	1 million characters (500 pages of text)
1 Gigabyte	1 billion characters (500,000 pages of text)
1 Terabyte	1 trillion characters (1,000,000 thick books)



Hard disk drives can store terabytes of data. This is the equivalent of many libraries full of books.



[Previous - Ways in which Data is Stored and Read](#)

[Next - Magnetic Storage Media](#)



Links to Theory Units:

[Section 1: Types and Components of Computer Systems](#)

[Section 4: Computer Networks](#)

[Section 7: The Ways in which ICT is used](#)

[Section 2: Input and Output Devices](#)

[Section 5: Data Types](#)

[Section 7: Part 2](#)

[Section 3: Storage Devices and Media](#)

[Section 6: The Effects of Using ICT](#)

[Section 8: Systems Analysis and Design](#)

Links to Practical Units:

[Section 9: Communication](#)

[Section 12: Integration](#)

[Section 15: Web site Authoring](#)

[Back to top](#)

[Section 10: Document Production](#)

[Section 13: Output Data](#)

[Section 16: Presentation Authoring](#)

[Section 11: Data Manipulation](#)

[Section 14: Data Analysis](#)